

REMEMBER FORWARD

BUILD A TIME CAPSULE FOR CIVILIZATION

Not for a school project. For the actual future of humanity.

THE STANDARD HOUSEHOLD CAPSULE · \$75–\$150 · HDPE PIPE

What Changed In This Edition

This edition replaces the Schedule 80 PVC design issued through August 2026. PVC has been withdrawn from the project.

PVC degrades by dehydrochlorination: it releases hydrogen chloride gas, the released acid accelerates further degradation, and inside a sealed capsule there is nowhere for that acid to go except onto your contents. PVC has also only existed since the 1930s, so the 500–1,000 year figure previously printed here had no evidence behind it at all.

HDPE is a plain hydrocarbon — carbon and hydrogen, no chlorine, nothing acidic to release. The closure method changes with it, because HDPE does not solvent-weld the way PVC does. See 'Closing The Capsule'.

What Is This?

A time capsule containing laser-engraved titanium plates carrying what a surviving community would need to rebuild: how to make water safe, how to make fire and charcoal, how to grow food, how to organise fairly, and how to read the languages we wrote in. It is designed to be found by someone who shares no language and no culture with us.

The project is called Remember Forward. The plates are called The Patient Message. Everything is free to download, copy, engrave and bury. No permission needed.

Why HDPE

- No chlorine. Nothing in the polymer can turn into acid and attack the contents.
- Buy the BLACK pipe. The carbon black is not cosmetic — it is the antioxidant and UV stabiliser, and black pipe is the standard buried grade.
- DR-11 or thicker wall. Under soil pressure over centuries, wall thickness is what resists slow deformation.
- Honest expectation: a design target of 100+ years, far shorter than the titanium tiers. That is the point of this tier. Its job is to be cheap enough that there are thousands of them, not to be the one that lasts longest.

Materials

Item	Cost	Where to Find
4" black HDPE pipe, DR-11, 18–24" section	\$20–\$35	Irrigation or plumbing supplier
4" HDPE socket-fusion end caps (2)	\$15–\$25	Same supplier
Socket fusion iron — rent, borrow, or supplier-fused	\$0–\$60	Tool rental; many irrigation suppliers fuse on site
Silica gel desiccant packets (8–10)	\$5–\$10	Online or hardware store

Item	Cost	Where to Find
Oxygen absorbers (4–6 packets)	\$4–\$8	Online or hardware store
Mylar vacuum-seal bags	\$5–\$10	Online
Plate set — see note below	see errata	Free to download at rememberforward.org

Plate-set prices are STALE and under revision. They were calculated for nickel, and never included engraving setup, which is charged per unique design. Treat them as an order of magnitude only. See the errata.

Closing The Capsule

There is no thread, no gasket, no tape and no epoxy in this design. HDPE is welded to itself, and the capsule is closed permanently.

That is deliberate. Every re-openable seal is a permanent liability bought to serve a single event that will be destructive anyway — the finder will cut it open. A capsule with a 100-year gasket is a 100-year capsule no matter what it is made of. Sweden's KBS-3 nuclear canisters, designed for 100,000 years, use welded lids and no seal at all.

Method A — socket fusion (preferred)

- Heat the pipe end and the cap socket on the fusion iron until both surfaces are glossy — typically 200–220 °C, 8–12 seconds for 4" pipe.
- Push together to the stop in one movement. Do not twist.
- Hold 60 seconds without moving. Leave undisturbed 30 minutes to cool fully.
- Do not exceed about 250 °C. Above that HDPE degrades rather than welds.

Method B — hot plate

- Face both parts flat on a clean steel plate heated to 200–220 °C until a bead of melt appears around the edge.
- Press together squarely and hold until cool. A misaligned joint is a weak joint.

Test before you load it

- Fuse one cap. Fill the tube with water, stand it upright 24 hours, check for weeping at the joint.
- Dry the inside completely. Load. Then fuse the second cap.
- Better: submerge the finished capsule and press gently. Any bubble is a failed weld.

Burial Instructions

- Minimum depth 4 feet; 5 to 6 feet is better, and below the local frost line. Deeper than plough depth — cultivation reaches 8 to 12 inches.
- Orientation horizontal — this distributes soil pressure along the length rather than onto the welded ends.
- Surround with clean gravel before backfilling, for drainage.
- Mark with a COPPER or BRONZE stake driven 12 inches above the capsule. Earlier editions said "non-ferrous (copper or stainless steel)" — stainless steel is an iron alloy and is not non-ferrous. It will corrode and stain over this timescale.

- Trade-off worth knowing: a steel stake is far easier to find with a metal detector, but will not last. If findability matters more than the marker's own lifespan, set a steel plate ABOVE the capsule and accept that it is a century-scale marker, not a permanent one.
- Record GPS coordinates and a description by landmarks — distances from two prominent natural features — and store both in several places.

Free Downloads

All SVG plate files, build guides, and this document are free at rememberforward.org — CC BY-SA 4.0, no permission needed.

Remember Forward · The Patient Message

Buy it. Build it. Bury it. For someone you will never meet.

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